

TABLE I
Radio Spectra

Band name	Abbreviation	Frequency	Uses
Tremendously-low frequency	TLF	<3Hz	Natural and artificial electromagnetic noise
Extremely-low frequency	ELF	3Hz to 30Hz	Communication with submarines
Super-low frequency	SLF	30Hz to 300Hz	Communication with submarines
Ultra-low frequency	ULF	300Hz to 3000Hz	Communication with submarines and within mines
Very-low frequency	VLF	3kHz to 30kHz	Navigation, time signals, submarine communication, wireless heart rate monitors, geophysics
Low frequency	LF	30kHz to 300kHz	Navigation, clock time signals, AM long-wave broadcasting (Europe and parts of Asia), RFID, amateur radio
Medium frequency	MF	300kHz to 3000kHz	AM (medium-wave) broadcasts, amateur radio, avalanche beacons
High frequency	HF	3MHz to 30MHz	Shortwave broadcasts, citizens' band radio, amateur radio and over-the-horizon aviation communications, RFID, over-the-horizon radar, automatic link establishment (ALE)/near-vertical incidence sky-wave (NVIS) radio communications, marine and mobile radio telephony
Very-high frequency	VHF	30MHz to 300MHz	FM, television broadcasts and line-of-sight ground-to-aircraft and aircraft-to-aircraft communications, land mobile and maritime mobile communications, amateur radio, weather radio
Ultra-high frequency	UHF	300MHz to 3000MHz	Television broadcasts, microwave oven, microwave devices/communications, radio astronomy, mobile phones, wireless LAN, Bluetooth, ZigBee, GPS and two-way radios such as land mobile, FRS and GMRS radios, amateur radio
Super-high frequency	SHF	3GHz to 30GHz	Radio astronomy, microwave devices/communications, wireless LAN, most modern radars, communications satellites, satellite television broadcasting, DBS, amateur radio
Extremely-high frequency	EHF	30GHz to 300GHz	Radio astronomy, high-frequency microwave radio relay, microwave remote sensing, amateur radio, directed-energy weapon, millimeter wave scanner
Terahertz or tremendously-high frequency	THz or THF	300GHz to 3,000GHz	Terahertz imaging, a potential replacement for X-rays in some medical applications, ultra-fast molecular dynamics, condensed-matter physics, terahertz time-domain spectroscopy, terahertz computing/communications, sub-mm remote sensing, amateur radio

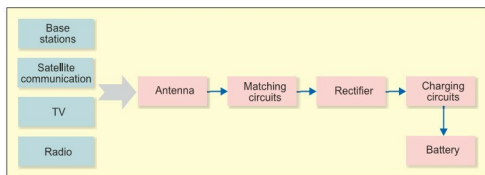


Fig. 2: Block diagram of an RF harvesting system

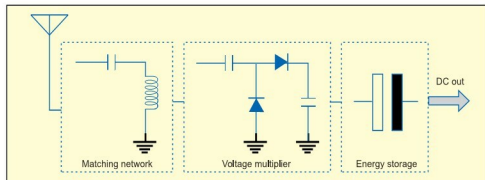


Fig. 3: Components of RF harvesting

multiplier. The matching network, composed of inductive and capacitive elements, ensures maximum power delivery from the antenna to the voltage multiplier. The energy storage ensures smooth power delivery to the load and as a reserve for durations when external energy is unavailable. Such a design needs to be carefully crafted; increasing the number of multiplier stages gives higher voltage at the load and reduces current through the final load branch. This may result in unacceptable charging delays for the energy-storage capacitor.

Conversely, fewer stages of the multiplier ensures quick charging of the capacitor, but voltage generated across it may be insufficient to drive the sensor mote (at least 1.8V that becomes +Vcc for Mica2 sensors).